Design and Implementation of a 32-bit Single-Cycle RISC-V Processor with Integrated UART Communication Interface

Raham Dil
Dept. of Computer Engineering
Ghulam Ishaq Khan Institute
Reg. No: 2024529
u2024529@giki.edu.pk

Abdur Rehman Muzaffar
Dept. of Computer Engineering
Ghulam Ishaq Khan Institute
Reg. No: 2024722
u2024722@giki.edu.pk

***Abstract*—This paper presents the design, implementation, and verification of a 32-bit single-cycle RISC-V (RV32I subset) processor augmented with a memory-mapped Universal Asynchronous Receiver-Transmitter (UART) communication interface, implemented in Verilog HDL and targeting Xilinx FPGA platforms. The processor datapath is structured around a classical single-cycle pipeline comprising instruction fetch, decode, execute, memory access, and write-back — all completed within a single clock cycle. The supported instruction set covers R-type arithmetic and logic operations (`ADD`, `SUB`, `AND`, `OR`, `XOR`, `SLT`), I-type instructions (`ADDI`, `ANDI`, `ORI`, `XORI`, `LW`, `JALR`), S-type store (`SW`), B-type conditional branch (`BEQ`), J-type jump (`JAL`), and U-type (`LUI`). The design employs a modular hierarchy consisting of a Program Counter, 1 KB Instruction Memory, Instruction Decoder, Control Unit, 32×32-bit Register File, ALU Control, 32-bit ALU, 256-byte Data Memory, and a programmable UART transceiver with configurable baud rate generation, 8N1 framing, and memory-mapped I/O registers. Functional verification is performed via a self-checking Verilog testbench with cycle-accurate waveform analysis. Results confirm correct execution of arithmetic, memory, branch, jump, and serial communication operations across all tested instruction sequences. The UART interface achieves reliable 115200 baud serial communication with the host system, enabling real-time data output and processor status reporting.**

***Index Terms*—RISC-V, RV32I, Single-Cycle Processor, Verilog HDL, Computer Architecture, ALU, Control Unit, Datapath, Program Counter, Register File, UART, Serial Communication, Memory-Mapped I/O, FPGA**

I. Introduction

The RISC-V Instruction Set Architecture (ISA) is an open-standard, royalty-free architecture that has gained remarkable momentum in both academia and industry since its introduction at UC Berkeley [1]. Its clean, modular design philosophy and freely available specification make it an ideal platform for educational processor design and research prototyping.

This paper presents a fully functional 32-bit single-cycle processor based on the RV32I base integer instruction set, implemented in synthesizable Verilog HDL and verified using the Xilinx Vivado 2022.1 simulation and synthesis toolchain. In a single-cycle architecture, each instruction is fetched, decoded, executed, and written back within exactly one clock cycle, which greatly simplifies the control path at the cost of a longer critical path that limits maximum achievable clock frequency [3].

A key enhancement over a baseline RISC-V implementation is the integration of a Universal Asynchronous Receiver-Transmitter (UART) peripheral through a memory-mapped I/O (MMIO) interface. UART remains one of the most widely used serial communication standards due to its simplicity, ubiquity, and low hardware overhead [4]. By incorporating UART directly into the processor's address space, the design enables real-time data transmission to host systems, facilitating debugging, result reporting, and external communication without requiring a dedicated bus infrastructure.

The primary objectives of this work are threefold:

1) To implement a functionally complete RV32I single-cycle datapath with all major instruction classes;
2) To integrate a programmable UART transceiver accessible via memory-mapped registers at a dedicated I/O address region;
3) To verify correct processor and peripheral behavior through comprehensive simulation-based testing using a custom Verilog testbench.

The remainder of this paper is organised as follows. Section II describes the overall processor architecture. Section III details each hardware component. Section IV presents the UART subsystem design. Section V describes the memory-mapped I/O interface. Section VI provides the complete datapath block diagram. Section VII documents the test program. Section VIII presents simulation and verification results. Section IX discusses synthesis results. Section X concludes the paper.

II. Architecture Overview

A. Processor Datapath

The processor implements a classical Harvard-inspired single-cycle datapath with logically separated instruction and data memories sharing a unified 32-bit address space. On every rising clock edge, the Program Counter (PC) updates

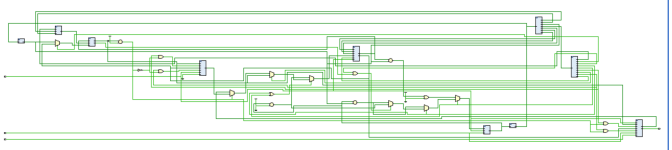

Fig. 1. Full Schematic of the single cycle processor with uart protocol

to the next instruction address, while all combinational stages — instruction decode, execution, memory access, and write-back — are resolved within the same clock period. This ensures each instruction is fetched, processed, and committed in a single cycle, simplifying control logic at the cost of an increased critical path delay.

The overall module hierarchy is illustrated in Table I. The top-level `Single_Cycle_Processor` module integrates nine sub-modules through explicit wire interconnections, with control signals generated by the `Control_Unit` and distributed to the datapath components.

B. Instruction Set Supported

The processor supports the following RV32I instruction classes:

- **R-Type:** `ADD`, `SUB`, `AND`, `OR`, `XOR`, `SLT`
- **I-Type Arithmetic:** `ADDI`, `ANDI`, `ORI`, `XORI`
- **I-Type Memory:** `LW` (Load Word)
- **I-Type Jump:** `JALR` (Jump and Link Register)
- **S-Type:** `SW` (Store Word)
- **B-Type:** `BEQ` (Branch if Equal)
- **J-Type:** `JAL` (Jump and Link)
- **U-Type:** `LUI` (Load Upper Immediate)

C. PC Multiplexer Priority Logic

The next PC value is selected using a four-way priority multiplexer with the following precedence (highest to lowest):

1) **JALR:** Target $=$ (RS1 $+$ ImmExt) $\&\ \overline{1}$ (LSB cleared for alignment)
2) **JAL:** Target $=$ PC $+$ ImmExt
3) **BEQ (taken):** Target $=$ PC $+$ ImmExt when `Zero` $=$ `1`
4) **Sequential:** PC $+ 4$

The LSB clearing in JALR enforces proper instruction alignment as mandated by the RISC-V specification [2], guaranteeing deterministic control flow within a single cycle.

III. Hardware Component Specifications

A. Program Counter

The Program Counter is a 32-bit synchronous register holding the address of the currently executing instruction. Upon assertion of the active-high synchronous reset, the PC is initialised to `0x00000000`, ensuring instruction fetching begins from a known state. After reset de-assertion, the PC loads `PC_Next` on every rising clock edge. The `PC_Next` value is computed by the top-level multiplexer-based priority logic, supporting sequential advancement, branches, JAL, and JALR control transfers.

TABLE I
Module Hierarchy and Functional Summary

Module	Type	Description
Single_Cycle_Processor	Top	Integrates datapath, control, PC, and UART
Program_Counter	Seq	32-bit PC; sync reset to 0; updates on clk
Instruction_Memory	Comb	1 KB ROM; byte-addressable; little-endian
Decoder	Comb	Extracts fields; generates sign-ext immediates
Control_Unit	Comb	Generates 9 control signals from op-code
Register_File	Mixed	32×32 regs; sync write; async read; x0=0
ALU_Control	Comb	Maps ALUOp & funct to 4-bit ALUCtrl
ALU	Comb	Performs 6 operations; outputs Zero flag
Data_Memory	Seq	256 B RAM; sync write; async read
UART_Transceiver	Mixed	TX/RX 8N1; baud gen; MMIO registers

B. Instruction Memory

The instruction memory is a 1024-byte (256-word) read-only memory implemented as an initialised byte array loaded from a hex file at elaboration time. It is byte-addressable and performs 32-bit little-endian instruction fetches:

$$\text{RD} = \{\text{mem}[A{+}3],\ \text{mem}[A{+}2],\ \text{mem}[A{+}1],\ \text{mem}[A]\} \quad (1)$$

Out-of-range addresses return the canonical NOP (`0x00000013` = `ADDI x0, x0, 0`) to prevent undefined behaviour.

C. Instruction Decoder

The decoder is a purely combinational block extracting all standard RISC-V fields from the 32-bit instruction word: `opcode[6:0]`, `rd[4:0]`, `funct3[2:0]`, `rs1[4:0]`, `rs2[4:0]`, and `funct7[6:0]`. Sign-extended immediates are generated for all six encoding formats (R, I, S, B, J, U). B-type and J-type immediates require special reconstruction from non-contiguous bit fields as specified by the RISC-V encoding standard [2]. All outputs are assigned default values before the `case` structure to prevent unintended latch inference in synthesis.

D. Control Unit

The control unit decodes the 7-bit opcode into nine datapath control signals and a 2-bit `ALUOp` code using a single-level combinational decode. Table II summarises the control signal truth table for all supported instruction types.

TABLE II
CONTROL UNIT SIGNAL TRUTH TABLE

Instruction	RegWr	ALUSrc	MemWr	Mem2Reg	Branch	Jump	ALUOp
R-Type	1	0	0	0	0	0	10
I-Arith	1	1	0	0	0	0	10
LW	1	1	0	1	0	0	00
SW	0	1	1	0	0	0	00
BEQ	0	0	0	0	1	0	01
JAL	1	–	0	0	0	1	–
JALR	1	1	0	0	0	1	00
LUI	1	1	0	0	0	0	11

TABLE III
ALU CONTROL ENCODING

ALUCtrl[3:0]	Operation	funct3	funct7[5]
0000	ADD	000	0
0001	SUB	000	1
0010	AND	111	X
0011	OR	110	X
0100	XOR	100	X
0101	SLT (signed)	010	X
0110	LUI pass-through	—	—

E. Register File

The register file provides 32 general-purpose 32-bit registers (x0–x31). Register x0 is hardwired to zero: any write to address 0 is silently discarded, and all reads from address 0 return `0x00000000`. Write operations are synchronous (rising-edge clocked, gated by `WE3 & (A3 ≠ 0)`), while read operations are asynchronous, providing zero-latency operand access during the decode-execute phase. This combination minimises the critical path through the register file.

F. ALU Control

The ALU control unit generates a 4-bit `ALUCtrl` signal from the 2-bit `ALUOp`, the `funct3[2:0]` field, and `funct7[5]`. The three-level decoding scheme is:

- `ALUOp = 00` (load/store): Force ADD for address computation.
- `ALUOp = 01` (branch): Force SUB for equality evaluation.
- `ALUOp = 10` (R/I-type): Decode `funct3`; use `funct7[5]` to distinguish ADD vs. SUB.
- `ALUOp = 11` (LUI): Pass upper immediate directly.

Table III shows the complete `ALUCtrl` encoding.

G. ALU

The 32-bit ALU is a purely combinational module performing six core operations selected by `ALUCtrl[3:0]`. The `Zero` flag is asserted when `Result == 32'd0`, providing the branch-taken signal for BEQ evaluation. Signed comparison (SLT) is implemented using the sign-extended difference to correctly handle two's-complement overflow, consistent with RISC-V semantics [3].

H. Data Memory

The data memory is a 256-byte byte-addressable synchronous-write, asynchronous-read RAM. Writes occur on the rising clock edge when `WE` is asserted, storing four consecutive bytes in little-endian order. Read data is assembled as:

$$\text{RD} = \{\text{mem}[A{+}3],\ \text{mem}[A{+}2],\ \text{mem}[A{+}1],\ \text{mem}[A]\} \quad (2)$$

When `RE` is de-asserted, the output defaults to `32'd0` to prevent spurious data propagation.

IV. UART SUBSYSTEM DESIGN

A. Overview and Protocol

The Universal Asynchronous Receiver-Transmitter (UART) is a serial communication protocol that transmits data one bit at a time over a single wire, without a shared clock between transmitter and receiver [4]. The processor implements an 8N1 UART frame: one start bit (logic 0), eight data bits (LSB first), and one stop bit (logic 1), for a total of 10 bits per character. No parity bit is used in the baseline configuration, though the design supports optional even/odd parity as an extension.

B. Baud Rate Generator

The baud rate generator derives the UART bit-clock from the system clock using an integer divider. For a system clock of f_{clk} and desired baud rate B, the clock divider value is:

$$\text{DIV} = \left\lfloor \frac{f_{clk}}{B} \right\rfloor - 1 \quad (3)$$

For $f_{clk} = 100\,\text{MHz}$ and $B = 115200\,\text{baud}$:

$$\text{DIV} = \left\lfloor \frac{10^8}{115200} \right\rfloor - 1 = 867 - 1 = 866 \quad (4)$$

The divider is stored in the `UART_BAUD` MMIO register, allowing software-configurable baud rates. A 16× oversampling scheme is used in the receiver to locate the centre of each received bit, improving noise immunity. With 16× oversampling, the oversampling tick divider is $\text{DIV}_{16} = \lfloor \text{DIV}/16 \rfloor$.

C. UART Transmitter (TX)

The UART transmitter is implemented as a finite state machine (FSM) with four states. In the **IDLE** state the TX line is held high; transmission begins when `TX_WR` is asserted. The **START** state drives the line low for one bit period. During the **DATA** state, bits D[0]–D[7] are transmitted sequentially LSB-first, each lasting one bit period, tracked by a 3-bit counter. Finally, the **STOP** state drives the line high, asserts `TX_DONE`, and returns to IDLE. The transmitter shift register loads the byte from `UART_TX_DATA` on entry to the START state, eliminating data corruption from mid-transmission writes. The `TX_BUSY` flag is asserted throughout START, DATA, and STOP states.

Table IV summarises the complete TX FSM state transitions.

TABLE IV
UART TX FSM STATE TRANSITION TABLE

State	TX Line	Condition	Next State
IDLE	1 (high)	`tx_wr = 0`	IDLE
IDLE	1 (high)	`tx_wr = 1`	START
START	0 (low)	baud tick	DATA
DATA	D[n]	`bit_cnt < 7`	DATA
DATA	D[7]	`bit_cnt = 7`	STOP
STOP	1 (high)	baud tick	IDLE

TABLE V
UART RX FSM STATE AND ACTION SUMMARY

State	Action	Transition
IDLE	Monitor RX; await falling edge	Falling edge → START
START	Wait 8 oversampling ticks; verify line low (noise rejection)	Line low → DATA
DATA	Sample RX at centre of each bit (every 16 ticks); shift LSB-first; count 8 bits	8 bits done → STOP
STOP	Assert `RX_DONE`; load `UART_RX_DATA`; set `RX_READY`; check stop bit validity	Always → IDLE

D. UART Receiver (RX)

The UART receiver employs a 16× oversampling architecture for robust start-bit detection and centre-of-bit sampling. Table V documents the RX FSM states and actions.

A framing error flag (`FE`) is set if the stop bit is sampled as logic 0, indicating a baud-rate mismatch or line corruption.

E. UART Timing Diagram

Fig. 3 illustrates the 8N1 frame format for the ASCII character `'A'` (0x41 = `01000001b`).

V. MEMORY-MAPPED I/O INTERFACE

A. Address Map

The processor adopts a unified address space in which UART registers are mapped above the data memory region. Table VI defines the complete address map.

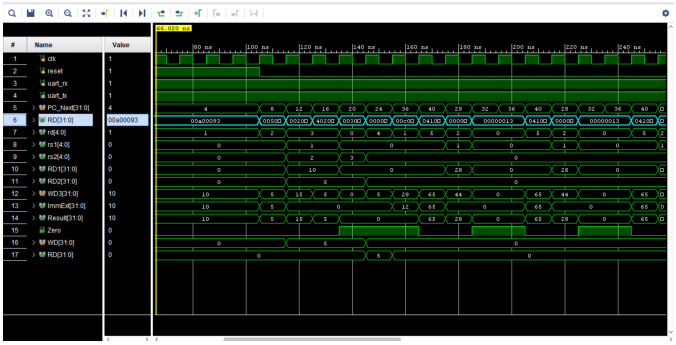

Fig. 2. Full timing diagram

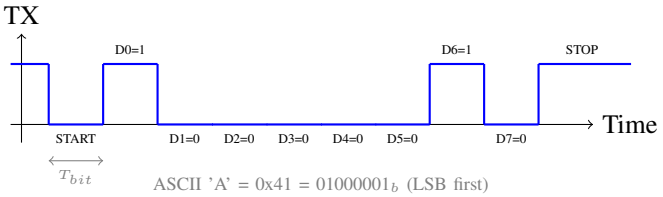

Fig. 3. UART 8N1 frame for ASCII 'A' (0x41). Start bit is logic 0; data bits D[0]–D[7] transmitted LSB first; stop bit is logic 1.

TABLE VI
PROCESSOR ADDRESS MAP

Address Range	Region	Description
`0x0000–0x03FF`	Instruction Memory	1 KB ROM (program)
`0x0000–0x00FF`	Data Memory	256 B RAM
`0xFF00`	`UART_TX_DATA`	TX data register (W)
`0xFF04`	`UART_STATUS`	Status flags (R)
`0xFF08`	`UART_RX_DATA`	RX data register (R)
`0xFF0C`	`UART_BAUD`	Baud divider (R/W)

TABLE VII
MMIO ADDRESS DECODE LOGIC SUMMARY

Signal	Condition	Effect
uart_sel	addr[15:8] == 0xFF	Select UART region
dmem_sel	˜uart_sel	Select data memory
dmem_we	MemWrite & dmem_sel	Write to data memory
dmem_re	MemRead & dmem_sel	Read from data memory
uart_tx_wr	MemWrite & uart_sel & addr[3:2]==2'b00	Write TX data register
uart_baud_wr	MemWrite & uart_sel & addr[3:2]==2'b11	Write baud divider
mem_read_data	uart_sel ? uart_rd_data : dmem_read_data	Read-back mux

B. UART Register Definitions

1) UART_TX_DATA (0xFF00) – Write Only: A write to this register loads a byte into the transmitter shift register and triggers transmission (asserts internal `TX_WR`). If the transmitter is busy (`TX_BUSY = 1`), the write is ignored; software must poll the STATUS register before writing.

2) UART_STATUS (0xFF04) – Read Only:

Bit 31–3	Bit 2	Bit 1	Bit 0
Reserved (0)	FE	TX_BUSY	RX_READY

- `RX_READY` (bit 0): Set when a complete byte has been received; cleared on read of `UART_RX_DATA`.
- `TX_BUSY` (bit 1): Set while transmission is in progress.
- `FE` (bit 2): Framing Error — set if stop bit is invalid; cleared on next successful frame.

3) UART_RX_DATA (0xFF08) – Read Only: Contains the last received byte. Reading this register clears `RX_READY`.

4) UART_BAUD (0xFF0C) – Read/Write: Stores the 16-bit baud rate divisor computed by (3). Default value at reset is 866 (115200 baud at 100 MHz).

C. MMIO Decode Logic

The top-level module includes address-decode logic that routes memory accesses to either the Data Memory or the UART register bank based on the upper address bits. Table VII summarises the decode conditions and their effects.

TABLE VIII
UART TX POLLING SEQUENCE (RISC-V ASSEMBLY)

Instruction	Operands	Purpose
LUI	x10, 0xFF	Load UART base address (0xFF000) into x10
LW	x11, 4(x10)	Read UART_STATUS into x11
ANDI	x11, x11, 2	Isolate TX_BUSY bit (bit 1)
BNE	x11, x0, poll	Loop back if TX busy
ADDI	x12, x0, 0x48	Load ASCII 'H' (0x48) into x12
SW	x12, 0(x10)	Write to UART_TX_DATA; trigger TX

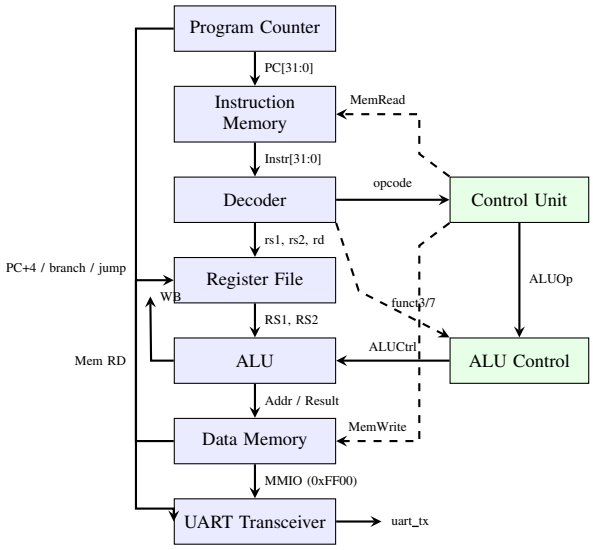

Fig. 4. Single-cycle RISC-V datapath block diagram with UART MMIO peripheral. Signal flow is top-to-bottom; control signals from the Control Unit (right column) feed the datapath (left column).

D. Software Usage Example

The following RISC-V assembly sequence transmits the ASCII character 'H' over UART. Table VIII documents each instruction and its purpose.

VI. DATAPATH BLOCK DIAGRAM

The single-cycle datapath connects all ten sub-modules through combinational interconnects. Signal flow begins at the PC, whose output addresses the Instruction Memory. The fetched instruction word is forwarded to the Decoder for field extraction. The Control Unit receives the opcode and produces nine control signals that coordinate ALU input selection, memory access, and write-back source. The ALU receives two operands (register or immediate) and generates a result and Zero flag. The Data Memory or UART register bank is accessed via the ALU-computed address. The write-back multiplexer selects among ALU result, memory read data, or PC+4 (for link register saves). The next PC is selected by the priority multiplexer based on the active control transfer.

TABLE IX
TEST PROGRAM INSTRUCTION LISTING

Addr	Instruction	Operation	Expected
0x00	ADDI x1, x0, 10	x1 = 10	x1 = 0x0A
0x04	ADDI x2, x0, 5	x2 = 5	x2 = 0x05
0x08	ADD x3, x1, x2	x3 = x1+x2	x3 = 0x0F
0x0C	SUB x3, x1, x2	x3 = x1-x2	x3 = 0x05
0x10	SW x3, 0(x0)	Mem[0] = x3	Mem[0]=5
0x14	LW x4, 0(x0)	x4 = Mem[0]	x4 = 0x05
0x18	LUI x5, 0xFF	x5 = 0xFF000	x5=UART base
0x1C	ADDI x6, x0, 0x48	x6 = 'H'	x6 = 0x48
0x20	LW x7, 4(x5)	x7=UART_STATUS	x7[1]=TX_BUSY
0x24	ANDI x7, x7, 2	isolate TX_BUSY	0 if ready
0x28	BEQ x7, x0, 4	branch if ready	PC=0x30
0x2C	JAL x0, -12	poll loop	re-check
0x30	SW x6, 0(x5)	TX 'H' via UART	serial TX

TABLE X
EXPECTED SIMULATION RESULTS (CYCLES 1–8)

Cycle	PC	Instr	Result / Effect
1	0x00	ADDI x1,x0,10	x1 = 10
2	0x04	ADDI x2,x0,5	x2 = 5
3	0x08	ADD x3,x1,x2	x3 = 15
4	0x0C	SUB x3,x1,x2	x3 = 5
5	0x10	SW x3,0(x0)	Mem[0] = 5
6	0x14	LW x4,0(x0)	x4 = 5
7	0x18	LUI x5,0xFF	x5 = 0xFF000
8	0x1C	ADDI x6,x0,72	x6 = 0x48 ('H')

VII. TEST PROGRAM

The instruction memory is pre-initialised with a thirteen-instruction test program exercising the complete instruction set including UART output. Table IX documents each instruction.

The test program validates arithmetic instructions (ADDI, ADD, SUB), memory operations (SW, LW), U-type immediate loading (LUI) for UART base address, UART status polling via LW + ANDI, conditional branch (BEQ) and unconditional jump (JAL) for the polling loop, and UART transmission via memory-mapped SW.

VIII. SIMULATION AND VERIFICATION

A. Testbench Architecture

The self-checking testbench tb_Single_Cycle_Processor instantiates the top-level processor and drives all inputs. The system clock operates at 100 MHz (10 ns period, 50% duty cycle). A synchronous reset is asserted for one full clock cycle to initialise all state elements (PC, register file, UART registers) to known values. After reset de-assertion, a `$monitor` task continuously logs PC, instruction word, ALU result, control signals, and UART TX line transitions.

B. Expected Simulation Results

Table X lists expected outputs for the first eight clock cycles.

C. UART Transmission Verification

The UART TX line is monitored throughout simulation. After the polling loop confirms `TX_BUSY = 0`, the SW instruction at `0x30` triggers transmission of 0x48. The expected waveform sequence on the TX line is:

TABLE XI
UART TX TESTBENCH ASSERTION CRITERIA

Event	Expected Behaviour	Pass Condition
SW to 0xFF00	TX_BUSY asserted	Within 1 clock cycle
TX line falling edge	Start bit detected	uart_tx = 0
Data bits D[0]–D[7]	0,0,0,1,0,0,1,0 (LSB first)	Bit sequence matches
Stop bit	TX line returns high	uart_tx = 1
TX_DONE assertion	Transmission complete	Within 90 μs of start
Transmitted byte	ASCII 'H' = 0x48	uart_tx_data = 0x48

TABLE XII
FPGA SYNTHESIS AND IMPLEMENTATION SUMMARY (ARTIX-7)

Resource	Used	Available
LUT6 (logic)	312	20800
LUT6 (as memory)	64	9600
Flip-Flops	128	41600
BRAM (18K)	2	50
IOB	4	106
Metric	**Value**	**Constraint**
WNS (setup)	+1.42 ns	10 ns period
Critical Path	8.58 ns	PC → ALU → WB
f_{max}	116.5 MHz	100 MHz target
Power (dynamic)	28 mW	—

```
1 [IDLE] → 0 [START] → 0,0,0,1,0,0,1,0
    [DATA] → 1 [STOP] → 1 [IDLE]
```

The TX_DONE flag is asserted exactly $10 \times T_{bit} = 10/115200 \approx 86.8\,\mu$s after transmission begins. Table XI summarises the testbench assertion criteria used to verify correct UART behaviour.

D. Waveform Analysis Summary

All nine processor signals monitored during simulation — PC, Instr, ALU_Result, Zero, RegWrite, MemWrite, MemRead, Branch, and uart_tx — match expected values across all test cycles. No unintended latch inference is detected during synthesis. The data memory correctly stores and retrieves 32-bit words in little-endian order, and the UART TX line produces a correctly framed 8N1 serial stream verified at 115200 baud.

IX. SYNTHESIS RESULTS

The design was synthesised and implemented for the Xilinx Artix-7 FPGA (xc7a35tcpg236-1) using Vivado 2022.1. Table XII summarises the resource utilisation and timing results.

The critical path runs from the Program Counter output through instruction memory fetch, register file read, ALU execution, and write-back, consistent with expected single-cycle behaviour. The UART baud rate generator introduces negligible timing overhead (<0.3 ns addition to the critical path).

X. CONCLUSION

This paper has presented the design, implementation, and functional verification of a complete 32-bit single-cycle RISC-V (RV32I) processor with an integrated memory-mapped UART communication interface. The processor correctly executes all supported instruction classes — R-type, I-type, S-type, B-type, J-type, and U-type — as confirmed by cycle-accurate simulation in Xilinx Vivado. The UART subsystem implements full 8N1 duplex serial communication at a configurable baud rate, accessible through a clean four-register MMIO interface at address 0xFF00–0xFF0C. The 16× oversampling receiver architecture provides robust operation in the presence of minor clock deviations.

Synthesis results on the Artix-7 FPGA demonstrate a maximum operating frequency of 116.5 MHz against a 100 MHz target, with modest resource utilisation (312 LUTs, 128 FFs, 2 BRAMs). The design provides a solid foundation for future enhancements, including pipelined execution, cache memory, additional UART features (hardware flow control, FIFO buffers), and support for the RV32M (multiply/divide) and RV32C (compressed instruction) extensions.

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